

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	192365013

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I K.Dhilip/192365013 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

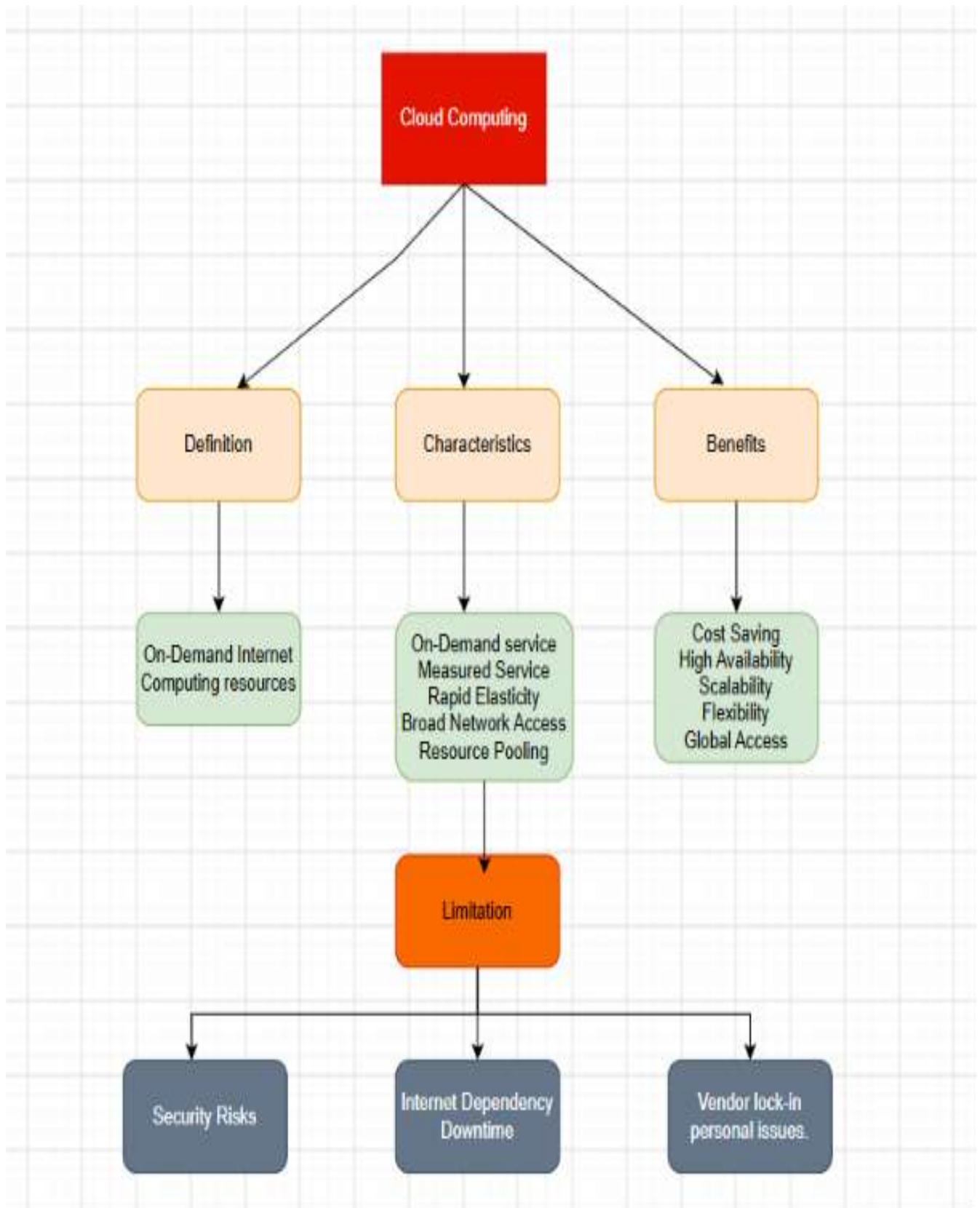
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

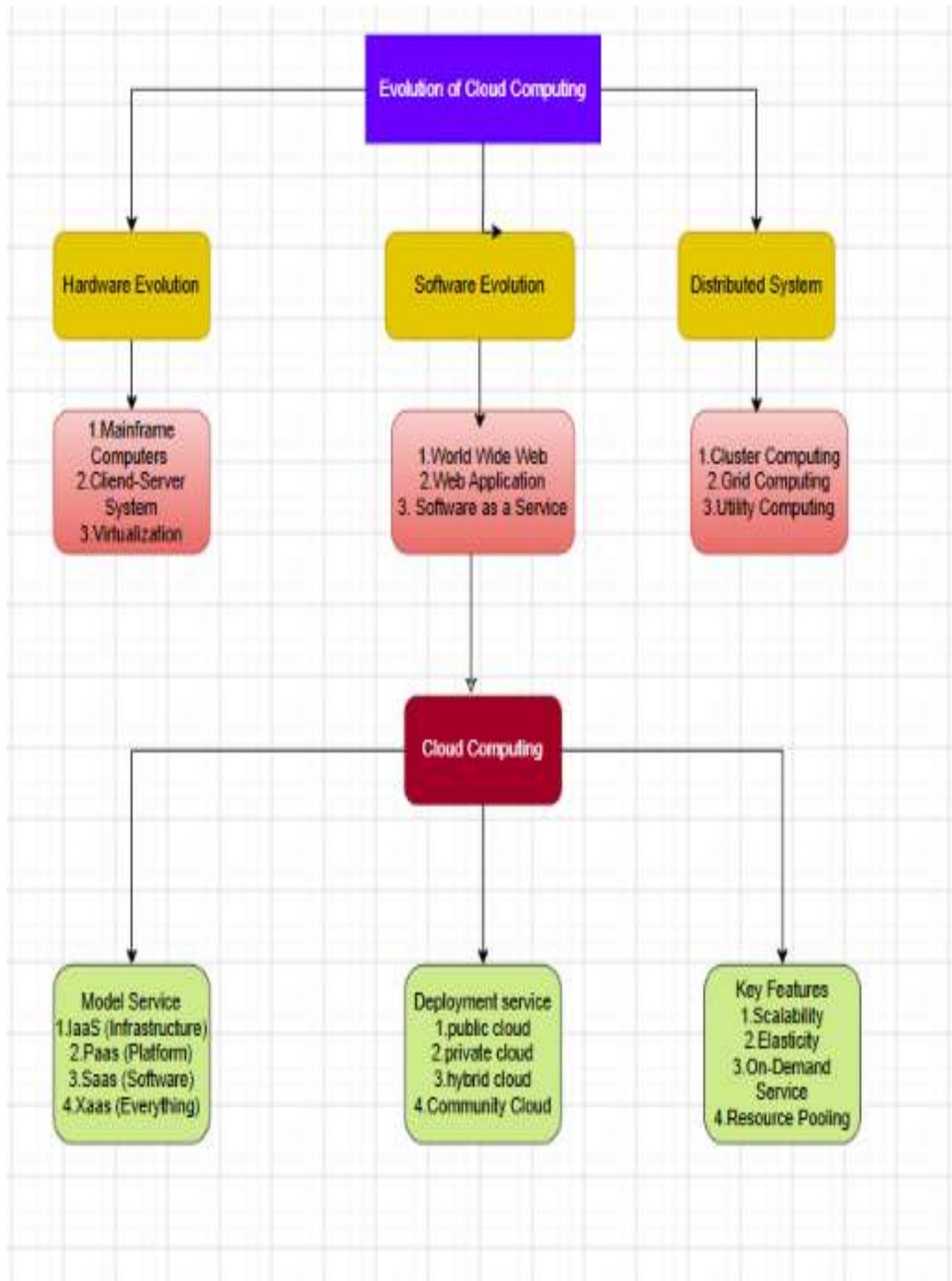
5. Questions

- i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
- ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
- iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
- iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
- v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

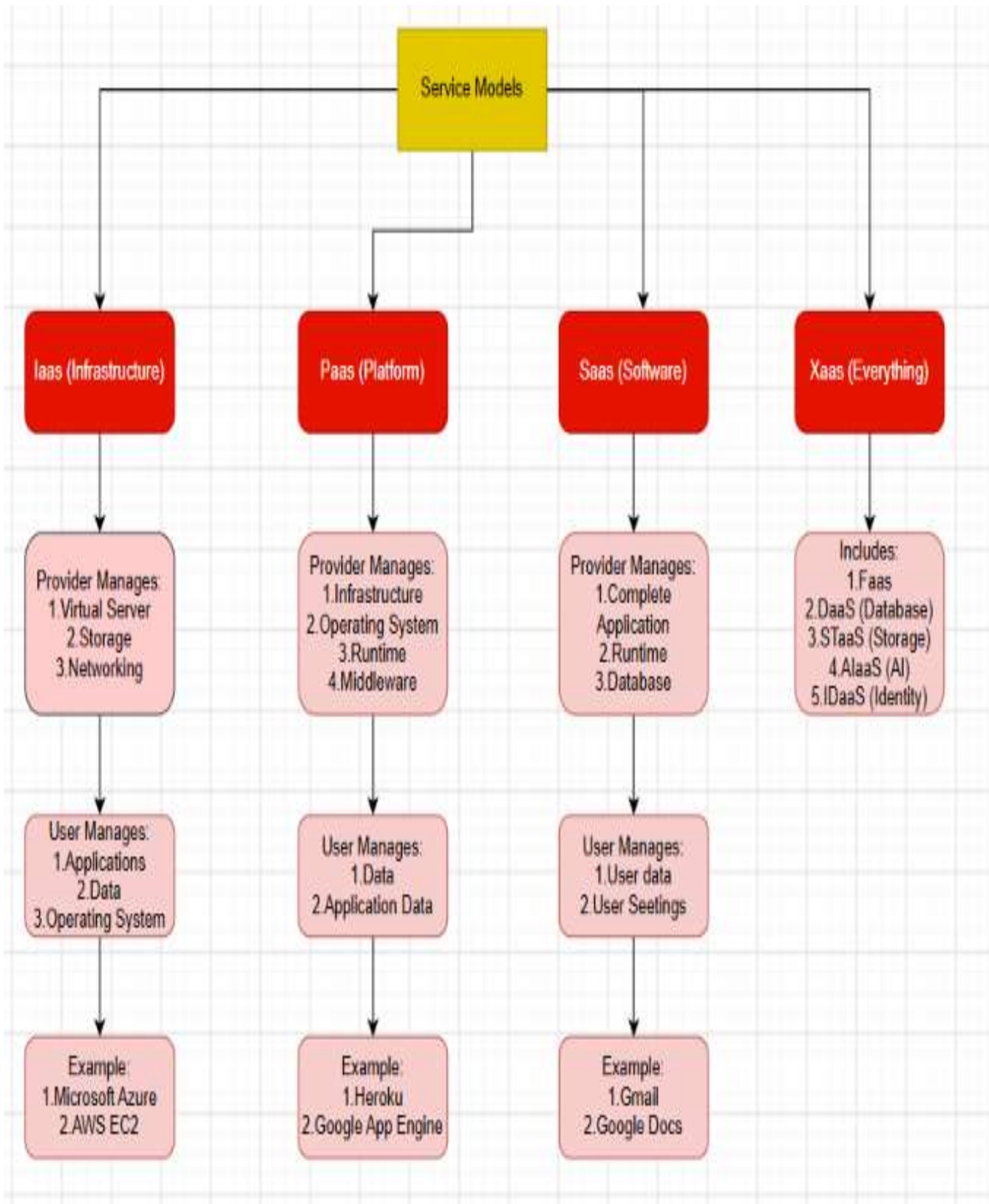
1. **Concept Map: Cloud Computing Fundamentals.** Include definition, characteristics, benefits, and limitations.



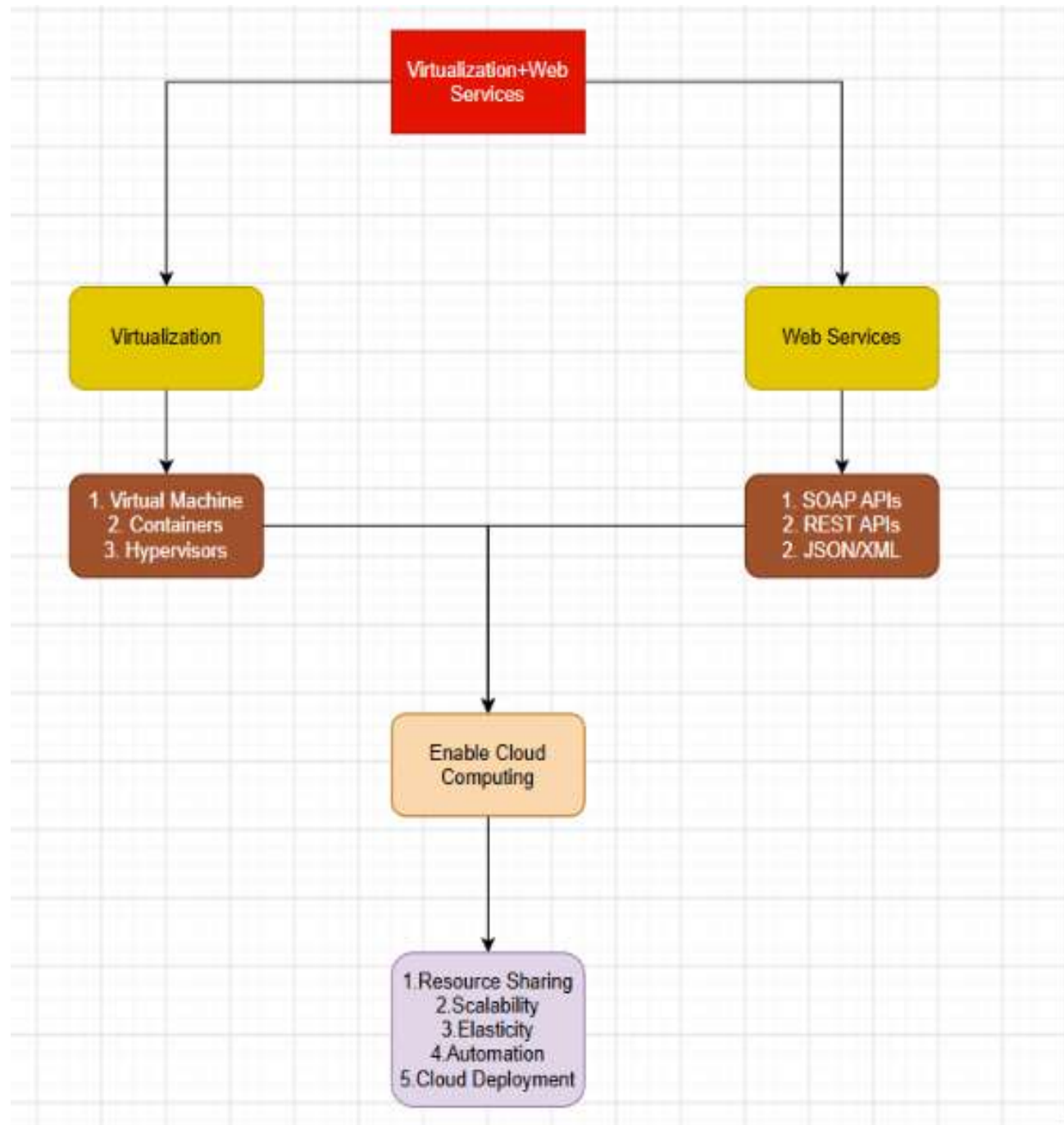
2. **Concept Map: Evolution of Cloud Computing.** Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



3. Concept Map: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



4. **Concept Map: Virtualization and Web Services as Enablers of Cloud.** Show how these two concepts support cloud deployment and scalability.



5.) i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

i.) The concept map effectively represents the core principles of cloud computing by organizing the concepts in a clear and logical manner.

ii.) It connects key topics such as service models, deployment models, virtualization, APIs, scalability, and elasticity. A hierarchical structure makes the relationships easy to understand. Labeled arrows clearly explain how each component interacts with others.

iii.) Colour coding improves readability and highlights different categories. Including real-world cloud services makes the map more practical and relevant. Overall, the design helps in understanding cloud architecture quickly and accurately.

ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

i.) Scalability allows the system to increase resources when user demand grows, ensuring smooth performance. Elasticity automatically adds or removes resources based on workload, helping to reduce costs.

ii.) Resource management efficiently allocates CPU, memory, storage, and network resources. Together, these features improve system performance, availability, and reliability. They also help applications handle traffic spikes without interruption.

iii.) Without these capabilities, the system may experience slow performance, downtime, resource wastage, and higher operational costs. This would negatively affect both the system's efficiency and the user experience.

iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

i.) The concept map includes REST APIs, web services, and cloud services because they are essential for building modern cloud applications. REST APIs enable fast and secure communication between the frontend and backend.

ii.) Web services allow different systems to exchange data seamlessly. Cloud services such as storage, databases, and authentication improve reliability and security. These services also support scalability and easy resource management.

iii.) They reduce development effort by providing managed infrastructure. Overall, they improve the functionality, performance, and efficiency of the entire system.

iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

i.) The components in the concept map are connected through APIs, cloud services, databases, and virtualization, enabling smooth communication between them. Authentication ensures secure access for users, while the backend processes requests efficiently.

ii.) The database and storage services provide reliable data management. Scalability and elasticity help the system handle varying workloads without affecting performance. Monitoring services detect issues and improve system reliability.

iii.) These interactions reduce response time, increase availability, and provide a fast, secure, and seamless user experience. Overall, the integrated architecture improves both performance and reliability.

v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

i.) Developing the concept map helped me understand how different cloud computing components are connected and work together. It improved my knowledge of service models, deployment models, APIs, virtualization, scalability, and elasticity.

ii.) Creating the map made complex concepts easier to visualize and remember. It also showed the importance of cloud services in building efficient applications. Analyzing the relationships improved my understanding of cloud architecture.

iii.) If I redesigned the map, I would include more security features, monitoring tools, and clearer labels. These improvements would make the concept map more detailed, organized, and easier to understand.